

Average Rate of Change

Logarithmic growth is characterized by a rate of change that decreases over time.

Finding the Rate of Change

A key feature for how a growth function behaves is the average rate of change from one point on the function to another.

Definition

The **average rate of change** of a function f from a to b is

$$\frac{f(b) - f(a)}{b - a}.$$

EXAMPLE 1

What is the average rate of change of the function $f(x) = \ln x + 1$ from 1 to 4?

SOLUTION

Compute $f(1)$ and $f(4)$.

$$f(1) = \ln 1 + 1 = 1$$

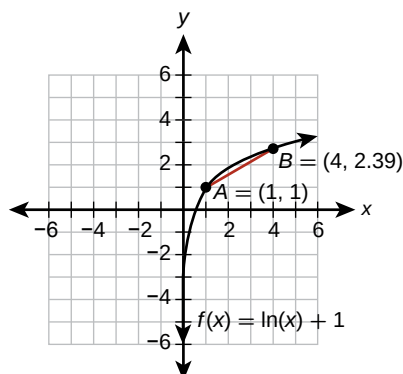
$$f(4) = \ln 4 + 1 \approx 2.39$$

Substitute the values into the formula for average rate of change.

$$\frac{f(b) - f(a)}{b - a} = \frac{2.39 - 1}{4 - 1} \approx 0.46$$

The average rate of change from 1 to 4 is about 0.46. ■

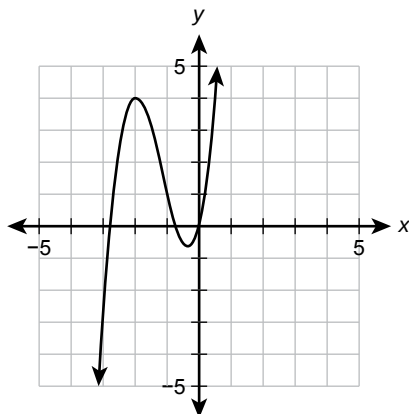
Visually, the average rate of change is the slope of the line connecting the two points.



► **Think About It** The average rate of change between any two points on a linear function is equal to its slope.

EXAMPLE 2

What is the average rate of change of the graphed function from $x = -2$ to $x = -1$?



SOLUTION

Find $f(-2)$ and $f(-1)$ using the graph.

$$f(-2) = 4$$

$$f(-1) = 1$$

Substitute the values into the formula for average rate of change.

$$\frac{f(b) - f(a)}{b - a} = \frac{1 - 4}{-1 - (-2)} = \frac{-3}{1} = -3$$

The average rate of change from $x = -2$ to $x = -1$ is -3 . ■

EXAMPLE 3

A local high school hired a new basketball coach in 2008. To determine her effectiveness as a coach, the school looked at the number of games the team won each season. The results are summarized in the table. What is the average rate of change from 2011 to 2014?

Year	2008	2009	2010	2011	2012	2013	2014
Number of games won	1	7	8	10	15	21	25

SOLUTION

Find $f(2011)$ and $f(2014)$ using the table.

$$f(2011) = 10$$

$$f(2014) = 25$$

Substitute the values into the formula for average rate of change.

$$\frac{f(b) - f(a)}{b - a} = \frac{25 - 10}{2014 - 2011} = \frac{15}{3} = 5$$

The average rate of change from 2011 to 2014 is 5. This means that, on average, the basketball team is winning 5 more games each season between 2011 and 2014. ■

Analyzing Logarithmic Growth

The average rate of change will be different at different parts of the function.

EXAMPLE 4

For the function $f(x) = \ln x + 1$, compare the average rate of change from 0.25 to 0.75, from 1.5 to 2, and from 7 to 7.5.

SOLUTION

Notice that each interval is a distance of 0.5 along the x -axis. Computing the function values gives $f(0.25) \approx -0.38$, $f(0.75) \approx 0.71$, $f(1.5) \approx 1.41$, $f(2) \approx 1.69$, $f(7) \approx 2.95$, and $f(7.5) \approx 3.01$. Use the function values to compute the average rates of change.

$$\text{from } 0.25 \text{ to } 0.75: \frac{0.71 - (-0.38)}{0.75 - 0.25} = 2.18$$

$$\text{from } 1.5 \text{ to } 2: \frac{1.69 - 1.41}{2 - 1.5} = 0.56$$

$$\text{from } 7 \text{ to } 7.5: \frac{3.01 - 2.95}{7.5 - 7} = 0.12$$

As the x -values increase, the average rate of change decreases. ■

In general, logarithmic growth means that the average rate of change decreases as x increases.